

Wireless Sensor Network Based Smart Healthcare Monitoring System

CS-314 Wireless Sensor Network
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Abstract—This report details the design, implementation, and evaluation of a Wireless Sensor Network (WSN) based Smart Healthcare Monitoring System. The system integrates wearable physiological and environmental sensors with a NodeMCU microcontroller to provide real-time monitoring and automated alert generation for patient safety. Key features, hardware and software components, data processing workflow, demonstration results, and future work are discussed.

Index Terms—Wireless Sensor Networks, Smart Healthcare, IoT, SpO₂, Temperature Monitoring, Real-time Alerts

I. INTRODUCTION

The rapid evolution of the Internet of Things (IoT) has enabled new paradigms in healthcare monitoring, especially for continuous patient tracking with minimal human intervention. Wearable and environmental sensors connected via WSNs can collect and transmit vital physiological data to centralized systems for analysis and alert generation. This project's primary objective is to design and implement a cost-effective, scalable, and energy-efficient system to monitor body temperature, blood oxygen saturation (SpO₂), heart rate, room temperature, and humidity in real time.

II. SYSTEM ARCHITECTURE

The proposed system consists of three major sensors:

- **Sensor Nodes:**

- Wearable sensors: DS18B20 for body temperature; MAX30100 for SpO₂ and heart rate.
- Environmental sensor: DHT22 for ambient temperature and humidity. Each sensor node periodically collects data and forwards it to the microcontroller unit.

- **Microcontroller Unit:**

- Interfaces with sensors over GPIO.
- Configures Wi-Fi connectivity to transmit timestamped JSON payloads.
- Implements threshold-based local decision logic for preliminary alerts.

- **Central Monitoring Interface:**

- A responsive web application developed with HTML, CSS, JavaScript, and Chart.js.
- Receives data via HTTP requests and displays live charts and widgets.
- Stores data locally with options for cloud integration to support historical analysis.

III. HARDWARE COMPONENTS AND CIRCUIT DESIGN

A detailed circuit diagram was created using an online simulator to validate sensor connections and power distribution. In the physical implementation, all sensors are powered by a regulated 5V source,

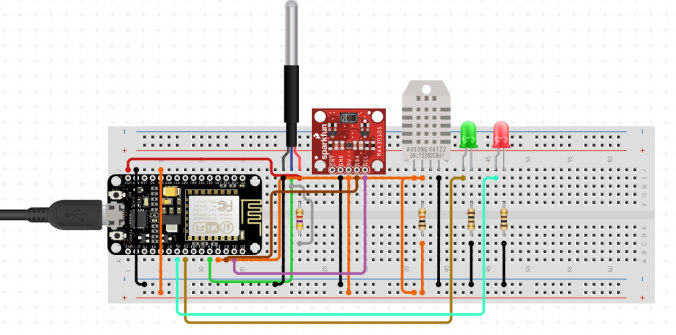


Fig. 1: Circuit schematic

with the NodeMCU operating at 3.3V logic levels.

- **DS18B20:** Digital temperature sensor for body temperature monitoring.
- **MAX30100:** SpO₂ and heart rate sensor.
- **DHT22:** Temperature and humidity sensor for ambient environmental monitoring.
- **NodeMCU:** ESP8266-based microcontroller with in-built Wi-Fi for data transmission.

IV. SOFTWARE DESIGN & IMPLEMENTATION

A. Firmware

- Developed in C++ (Arduino IDE); handles sensor initialization, calibration, and interrupt-driven sampling.
- Reads and packages sensor data into JSON every two seconds; implements Wi-Fi reconnection retries.

B. Web Dashboard

- Built with responsive UI; fetches data via AJAX for dynamic updates without reloads.
- Chart.js powers time-series plots; alert banners trigger on threshold breaches (e.g., SpO₂ less than 90%, temp greater than 38°C).

C. Data Workflow

- NodeMCU reads sensors → 2. Sends JSON POST to local server → 3. Server logs data → 4. Dashboard polls server → 5. Charts and widgets update in real time.

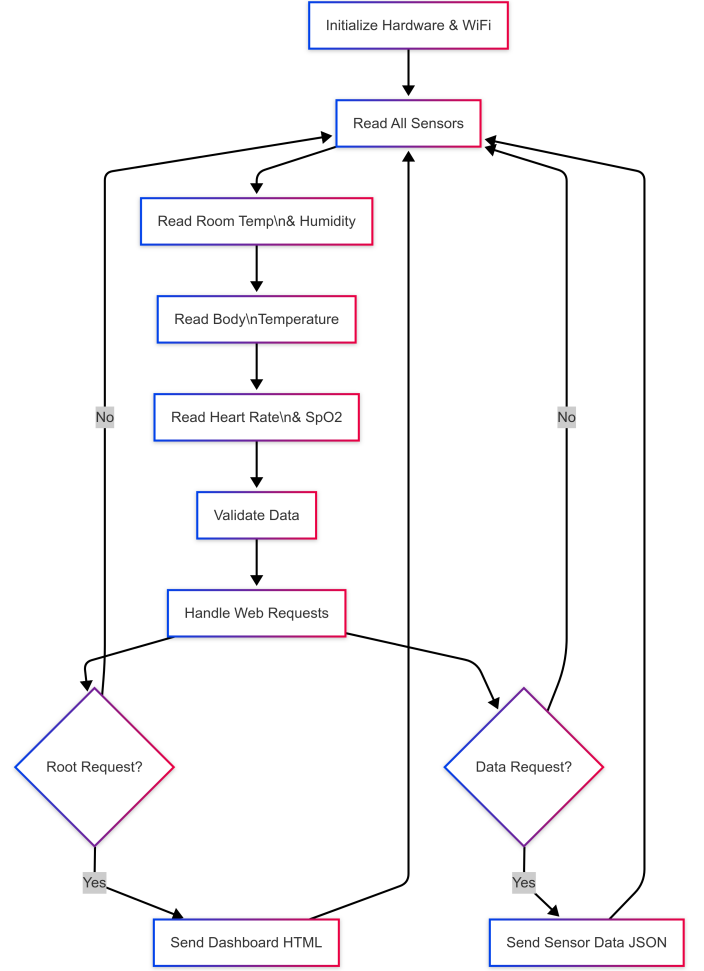


Fig. 2: Workflow Diagram

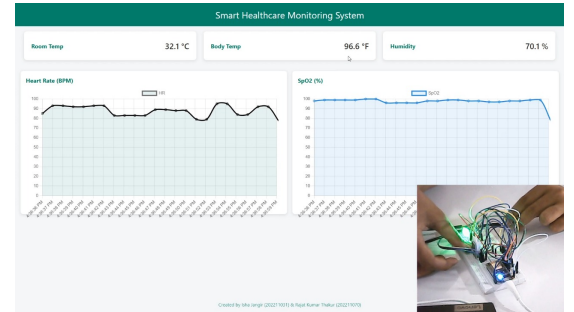


Fig. 3: Demonstration

V. DEMONSTRATION RESULTS

- Continuous real-time visualization of vital and environmental parameters.
- Automated alerts for critical conditions.
- Historical trend analysis.

VI. APPLICATIONS:

- **Elderly Monitoring:** Ensures round-the-clock care for elderly individuals in smart homes.
- **Post-Surgery Care:** Enables remote tracking of patient vitals during recovery periods.
- **Hospital Triage:** Automates continuous monitoring to alleviate medical staff workload.
- **Quarantine Zones:** Facilitates remote observation of infectious patients, reducing exposure risk.

VII. LIMITATIONS & CHALLENGES

- **Sensor Drift:** Requires periodic calibration to maintain accuracy.
- **Network Reliability:** Wi-Fi outages may interrupt data flow; MQTT could enhance resilience.
- **Power Constraints:** Battery-powered nodes need sleep cycles or energy harvesting for extended deployments.
- **Scalability:** Current design supports a single node; multi-node synchronization and address management are pending.

VIII. ETHICAL & DATA PRIVACY CONSIDERATIONS

- **Patient Consent:** Data collection complies with informed consent protocols.
- **Data Security:** Plans to implement TLS/SSL encryption and robust authentication to protect PII.
- **Privacy:** Anonymization of patient IDs in storage; adherence to relevant healthcare data regulations.

IX. FUTURE WORK

- **Anomaly Detection (ML):** Integrate server-side models for predictive alerts.
- **Multi-Patient Support:** Dashboard enhancements for concurrent monitoring with patient-specific tagging.

- **Energy Optimization:** Incorporate low-power modes and solar panels.
- **Smart IoT Integration:** Automate environment control via thermostats or smart beds.
- **Regulatory Compliance:** Align with HIPAA/GDPR standards for data handling.

X. CONCLUSION

This WSN-based Smart Healthcare Monitoring System demonstrates robust real-time monitoring, alerting, and trend analysis using off-the-shelf components and open-source technologies. Addressing current limitations and integrating advanced analytics and security measures will pave the way toward clinical-grade deployment.

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